Name

- Show appropriate work.
  - Appropriate work for a linear ODE includes the integrating factor and some intermediate steps.
  - Appropriate work for a missing eigenvector includes the linear system you are solving and the solution.
  - Labelling and using eigenvalues/eigenvectors etc. appropriately is a good idea.
- Explicitly real solutions are required.
- Many questions ask only for a linear system. Do not solve the system in these cases.
- Eigenvalues and eigenvectors are given. Do not attempt to recompute them.

Linear ODEs

Section 2.3

1. Solve the ODE $2\,y'(t) - 4\,y(t) = -6\,e^{2\,t}$.
2. Solve the ODE $2\,y'(t) - 4\,y(t) = -6\,e^{2\,t}$.
3. Solve the ODE $2\,y'(t) - 4\,y(t) = -6\,e^{2\,t}$ with $y(0) = 2$ (at end)
4. Solve the ODE $3\,y'(t) - 9\,y(t) = 6\,e^{3\,t}$.
5. Solve the ODE $3\,y'(t) + 6\,y(t) = -3\,e^{2\,t}$ with $y(0) = -3$
6. Solve the ODE $t\,y'(t) - 3\,y(t) = \frac{2}{t^3}$ (at end)
7. Solve the ODE $t\,y'(t) - 3\,y(t) = -3\,t^2$.
8. Solve the ODE $t\,y'(t) + 2\,y(t) = 2\,t^2$ with $y(1) = 2$ (at end)
9. Solve the ODE $t\,y'(t) + 2\,y(t) = 2\,t^2$.
10. Solve the ODE $t\,y'(t) + 3\,y(t) = 3\,t^3$ with $y(1) = -1$ (at end)

All with a box saying

$y(t) =$

Complementary Solutions from Eigenvalues/Eigenvectors

Sections 3.3, 3.4, & 3.5

Any and all of the questions from these sections can include. “Discuss the behavior as $t \to \infty$.”

1. Compute the complementary solution of $y' = A\,y$ where A is below

$$\texttt{MatrixForm}\left[\texttt{Eigensystem}\left[A = \begin{pmatrix} -5 & 6 & -9 \\ -3 & 4 & -9 \\ 0 & 0 & -2 \end{pmatrix}\right]\right]$$

λ_2 λ_3

Out[ ]//MatrixForm=

$$\begin{pmatrix} -2 & -2 & 1 \\ \{-3, 0, 1\} & \{2, 1, 0\} & \{1, 1, 0\} \end{pmatrix}$$

V_2 V_3

$$V_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\boxed{y_c(t) = c_1 e^{-2t}\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} + c_2 e^{\lambda_2 t} V_2 + c_3 e^{\lambda_3 t} V_3}$$

$e^{\lambda_3 t} v_3$ matters most. top 2 components go to ∞ like e^t.

2. Compute the complementary solution of $y' = A\,y$ where A is below

$$\texttt{MatrixForm}\left[\texttt{Eigensystem}\left[A = \begin{pmatrix} 3 & 0 & -5 \\ 5 & -2 & -5 \\ 0 & 0 & -2 \end{pmatrix}\right]\right]$$

Out[82]//MatrixForm=

$$\begin{pmatrix} 3 & -2 & -2 \\ \{1, 1, 0\} & \{1, 0, 1\} & \{0, 1, 0\} \end{pmatrix}$$

$\lambda_1 = 3$, $\quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$\lambda_2 = -2$, $\quad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$\lambda_3 = -2$, $\quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$$\boxed{y_c(t) = \sum_{i=1}^{3} c_i e^{\lambda_i t} v_i} \sim c_1 e^{3t}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \text{ as } t \to \infty$$

3. Compute the complementary solution of $y' = A\,y$ where A is below

$$\texttt{MatrixForm}\left[\texttt{Eigensystem}\left[A = \begin{pmatrix} 4 & 7 & -12 \\ 4 & 8 & -8 \\ 2 & 5 & -6 \end{pmatrix}\right]\right]$$

Out[101]//MatrixForm=

$$\begin{pmatrix} 6 & -2 & 2 \\ \{1, 2, 1\} & \{2, 0, 1\} & \{-1, 2, 1\} \end{pmatrix}$$

$\lambda_1 = 6$, $\quad v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

$\lambda_2 = -2$, $\quad v_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

$\lambda_3 = 2$, $\quad v_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$

$$\boxed{y_c(t) = \sum_{i=1}^{3} c_i e^{\lambda_i t} \vec{v}_i}$$

4. Compute the complementary solution of $y' = A\,y$ where A is below

$4+2i = a+bi$ $\quad a=4 \quad b=2$

$\begin{bmatrix} i \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$p = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$ $\quad q = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$y_3 = e^{-2t} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

```
MatrixForm[Eigensystem[A = (4 5 -12
                            4 8 -8
                            2 5 -6)]]
```

Out[105]//MatrixForm=

```
( 4 + 2 i      4 - 2 i      -2
  {i, 2, 1}  {-i, 2, 1}  {2, 0, 1} )
```

$y_1 = e^{at}\left(\cos(bt)\,p - \sin(bt)\,q\right)$

$y_2 = e^{at}\left(\sin(bt)\,p + \cos(bt)\,q\right)$

$y_c(t) = \sum_{i=1}^{3} c_i y_i(t)$

y_1 and y_2 get big and wiggle

5. Compute the complementary solution of $y' = A\,y$ where A is below

In[154]:=

```
MatrixForm[Eigensystem[A = (-6 -5  8
                             4 -2 -8
                             2  0 -6)]]
```

$y_3 = e^{-2t}\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

$y_1 = e^{-6t}\left(\cos(2t)\,p - \sin(2t)\,q\right)$

$y_2 = e^{-6t}\left(\sin(2t)\,p + \cos(2t)\,q\right)$

Out[154]//MatrixForm=

```
( -6 + 2 i    -6 - 2 i      -2
  {i, 2, 1}  {-i, 2, 1}  {2, 0, 1} )
```

$p = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$ $\quad q = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$y_c(t) = c_1 y_1 + c_2 y_2 + c_3 y_3$

y_3 most imp $t \to \infty$

solutions decay to zero like e^{-2t}

6. Compute the complementary solution of $y' = A\,y$ where A is below

$y_2 = t e^{3t}\begin{bmatrix} 1 \\ 2 \end{bmatrix} + e^{3t} u$

In[155]:=

```
MatrixForm[Eigensystem[A = (5 -1
                            4  1)]]
```

$y_1 = e^{3t} v_1$

$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$[A - \lambda_1 I]\,u = v_1$

Out[155]//MatrixForm=

```
(   3       3
  {1, 2}  {0, 0} )
```

$\lambda_1 = 3$

$\left[\begin{array}{cc|c} 5-3 & -1 & 1 \\ 4 & 1-3 & 2 \end{array}\right] = \left[\begin{array}{cc|c} 2 & -1 & 1 \\ 4 & -2 & 2 \end{array}\right] \sim \left[\begin{array}{cc|c} 2 & -1 & 1 \\ 0 & 0 & 0 \end{array}\right]$

$y_c(t) = c_1 e^{3t}\begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 y_2$

$u = \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}$

7. Compute the complementary solution of $y' = A\,y$ where A is below

$y_2 = t e^{4t}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{4t} u$

In[185]:=

```
MatrixForm[Eigensystem[A = (5 -1
                            1  3)]]
```

Out[185]//MatrixForm=

```
(   4       4
  {1, 1}  {0, 0} )
```

$[A - 4I]\,u = v$

$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 1 & -1 & 1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 0 & 0 \end{array}\right]$

$y_c(t) = c_1 e^{4t}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 y_2$

$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$c_2 y_2$ most important $\sim t e^{4t}$

Undetermined Coefficients

Section 3.6

1. For the ODE $y' = A\,y + b$ with $b = e^{3t}\begin{pmatrix}1\\3\end{pmatrix}$

$y_p' = 3ue^{3t}$

$3ue^{3t} = A(ue^{3t}) + e^{3t}\begin{pmatrix}1\\3\end{pmatrix}$

$3u = Au + \begin{pmatrix}1\\3\end{pmatrix}$

$(3I - A)u = \begin{pmatrix}1\\3\end{pmatrix}$

1.1. Write down the form for the particular solution

$y_p(t) = ue^{3t}$

1.2. Write down the Lin System for the coefficients

$\left[\begin{array}{c|c} 3I-A & \begin{matrix}1\\3\end{matrix}\end{array}\right]$

2. For the ODE $y' = A\,y + b$ with $b = \cos(4t)\begin{pmatrix}1\\3\end{pmatrix} + \sin(4t)\begin{pmatrix}4\\2\end{pmatrix}$

$y_p' = -4\sin(4t)u_1 + 4\cos(4t)u_2$

$= \cos(4t)Au_1 + \sin(4t)Au_2 + \cos(4t)\begin{pmatrix}1\\3\end{pmatrix} + \sin(4t)\begin{pmatrix}4\\2\end{pmatrix}$

$-4u_1 = Au_2 + \begin{pmatrix}4\\2\end{pmatrix}$

$4u_2 = Au_1 + \begin{pmatrix}1\\3\end{pmatrix}$

$-4Iu_1 - Au_2 = \begin{pmatrix}4\\2\end{pmatrix}$

$-Au_1 + 4Iu_2 = \begin{pmatrix}1\\3\end{pmatrix}$

2.1. Write down the form for the particular solution

$y_p(t) = \cos(4t)u_1 + \sin(4t)u_2$

2.2. Write down the Lin System for the coefficients

$\left[\begin{array}{cc|c} -4I & -A & \begin{matrix}4\\2\end{matrix} \\ -A & 4I & \begin{matrix}1\\3\end{matrix}\end{array}\right]$

3. For the ODE $y' = A\,y + b$ with $b = \begin{pmatrix}t^2\\1\\2\end{pmatrix}$

$y_p' = u_1 + 2tu_2$

$= Au_0 + tAu_1 + t^2Au_2 + \begin{pmatrix}0\\1\\2\end{pmatrix} + \begin{pmatrix}0\\0\\0\end{pmatrix}t + t^2\begin{pmatrix}1\\0\\0\end{pmatrix}$

$u_1 = Au_0 + \begin{pmatrix}0\\1\\2\end{pmatrix}$

$2u_2 = Au_1 + \begin{pmatrix}0\\0\\0\end{pmatrix}$

$0 = Au_2 + \begin{pmatrix}1\\0\\0\end{pmatrix}$

3.1. Write down the form for the particular solution

$y_p(t) = u_0 + tu_1 + t^2u_2$

3.2. Write down the Lin System for the coefficients

$\left[\begin{array}{ccc|c} -A & I & 0 & \begin{matrix}0\\1\\2\end{matrix} \\ 0 & -A & 2I & \begin{matrix}0\\0\\0\end{matrix} \\ 0 & 0 & -A & \begin{matrix}1\\0\\0\end{matrix}\end{array}\right]$

4. For the ODE $y' = A\,y + b$ with $b = \begin{pmatrix} e^{2t} \\ -e^{2t} \\ 3\,e^{2t} \end{pmatrix}$

4.1. Write down the form for the particular solution

$y_p(t) = u e^{2t}$

$$y_p' = 2u e^{2t} = e^{2t} A u + e^{2t} \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$$

$$2u = Au + \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$$

$$(2I - A)\, u = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$$

4.2. Write down the Lin System for the coefficients

$2u = Au + \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$

5. For the ODE $y' = A\,y + b$ with $b = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

5.1. Write down the form for the particular solution

$y_p(t) = u$

$$Au + b = 0$$

$$Au = -b$$

$$u = -A^{-1} b$$

5.2. Write down the Lin System for the coefficients

$Au = -b.$

#3

$$y' - 2y = -3e^{2t} \qquad y(0) = 2$$

$$v(t) = e^{-2t}$$

$$e^{-2t}y' - 2e^{2t}y = -3e^{-2t}e^{2t}$$

$$(e^{-2t}y)' = -3 \Rightarrow e^{-2t}y = \int -3\,dt + c$$

$$e^{-2t}y = -3t + c$$

$y(0)=2$ $\quad e^{-0}(2) = -3(0) + c$

$$\boxed{c = 2}$$

$$y = -3te^{2t} + ce^{2t}$$

$$\boxed{y = -3te^{2t} + 2e^{2t}}$$

#6 $y' - \frac{3}{t}y = 2t^{-4}$ (SF)

IF $v = e^{\int -\frac{3}{t}dt} = e^{-3\ln(t)} = t^{-3}$

$t^{-3}y' - 3t^{-4}y = 2t^{-7}$

$(t^{-3}y)' = 2t^{-7}$

$t^{-3}y = \int 2t^{-7}dt + C$

$t^{-3}y = \frac{2}{-6}t^{-6} + C$

$y = t^{3}\left(-\frac{1}{3}t^{-6} + C\right)$

Lin

#8 $ty' + 2y = 2t^2 \qquad y(1) = 2$

$y' + \frac{2}{t}y = 2t$

$v = e^{\int \frac{2}{t} dt} = t^2$

$t^2 y' + 2ty = 2t^3$

$(t^2 y)' = 2t^3$

$t^2 y = \int 2t^3 dt + c$

$t^2 y = 2t^4/4 + c$

$1^2(2) = 2\,1^4/4 + c$

$c = 2 - \frac{1}{2} = \frac{3}{2}$

$\boxed{y = \frac{2t^4}{4t^2} + c/t^2 = \frac{1}{2}t^2 + \frac{3}{2}t^{-2}}$

Lin

#10 $t y' + 3y = 3t^3$ $\boxed{y(1) = -1}$

$y' + \frac{3}{t} y = 3t^2$ Std Form

$v = e^{\int 3/t \, dt} = e^{3\ln(t)}$

$v = e^{\ln(t^3)} = t^3$

$t^3 y' + 3t^2 y = 3t^5$

$(t^3 y)' = 3t^5$

$t^3 y = \int 3t^5 \, dt + c$

$t^3 y = 3 \frac{t^6}{6} + c$

$y = \frac{3t^6}{6t^3} + ct^{-3}$

$y(1) = -1$ $-1 = \frac{3}{6} + c$

$\boxed{c = -1 - \frac{1}{2}} = -\frac{3}{2}$

$y = \frac{3t^6}{6t^3} - \frac{3}{2} t^{-3}$

Form and Lin Sys for y_p

$$y' = Ay + e^{2t}\begin{pmatrix}3\\4\end{pmatrix}$$

Form $\quad y_p = e^{2t}u$

$$y_p' = 2e^{2t}u$$

$$2\cancel{e^{2t}}u = A\cancel{e^{2t}}u + \cancel{e^{2t}}\begin{pmatrix}3\\4\end{pmatrix}$$

$$2u = Au + \begin{pmatrix}3\\4\end{pmatrix}$$

$$(2I - A)u = \begin{pmatrix}3\\4\end{pmatrix}$$

$$\left[\begin{array}{c|c} 2I - A & \begin{matrix}3\\4\end{matrix}\end{array}\right]$$